

REWARDLENS: SAME STATIC ACCURACY, DIFFERENT INTERVENTION BEHAVIOR IN MULTIMODAL JUDGES

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ABSTRACT

Static preference accuracy can fail to distinguish multimodal judges that behave differently under controlled changes in visual evidence. We introduce RewardLens, a behavioral audit that evaluates each judge on an independent static preference set and on a separate controlled intervention audit while holding the question and candidate responses fixed. Relevant edits are programmatically constructed and validated to change the correct choice, whereas irrelevant edits are constructed and validated to preserve it. Among audit cases judged correctly on the base image, RewardLens measures Relevant Adaptation (RA), correctness after a relevant edit, and Irrelevant Invariance (II), correctness after an irrelevant edit.

RewardLens asks whether an independent static score is sufficient to distinguish factor-specific intervention behavior. Across eight multimodal judges and four visual factors, equal or near-equal measured static accuracies can coexist with substantial differences in relevant adaptation. On Attribute, Phi-3.5-Vision and LLaVA-OneVision both achieve 80.5% measured static accuracy, yet differ in RA by 15.1 percentage points (paired-bootstrap 95% CI: 10.2–20.4 pp). Across seven within-factor model-pair comparisons whose measured static accuracies differ by at most one percentage point, the median RA gap is 15.1 points.

The difference is concrete: in Qwen’s Count audit, 166 of 200 base-correct triplets (83.0%) become incorrect after a relevant edit because the correct choice changes while the judge retains its original prediction. In these cases, invariance is the failure. RewardLens complements static preference accuracy by revealing differences in measured intervention behavior that conventional static evaluation can fail to distinguish.

1 INTRODUCTION

Multimodal judges compare vision–language outputs and provide preference signals for model development. They are commonly summarized by static preference accuracy: given an image, question, and two responses, how often is the correct response selected? This cross-sectional score does not reveal how the same decision changes when its visual evidence changes.

This question is adjacent to, but distinct from, controlled perturbation and failure-to-update studies (Lee et al., 2026; Park et al., 2026; Zhu et al., 2026). RewardLens evaluates the residual behavioral variation among judges that an *independent* static preference evaluation assigns equal or nearly equal scores. It also tests both a relevant edit, for which the gold preference must change, and an irrelevant edit, for which it must remain fixed.

We introduce **RewardLens**, a behavioral audit asking: *among judges with equal or near-equal measured static accuracy, how much intervention-behavior variation remains?* An independent natural-image set measures factor-specific static accuracy. A separate controlled audit then holds the question and responses fixed while changing the rendered scene. Each triplet contains a base image, a relevant edit that reverses the gold preference, and an irrelevant edit that preserves it. Relevant Adaptation (RA) measures correctness after the relevant edit, conditional on a correct base decision; Irrelevant Invariance (II) measures the analogous correctness after the irrelevant edit.

Across eight multimodal judges and four factors—Count, Attribute, Presence, and Spatial—static similarity can coexist with different measured intervention behavior. Phi-3.5-Vision and LLaVA-OneVision-7B tie at 80.5% static Attribute accuracy, yet differ in RA by 15.1 percentage points (paired-bootstrap 95% CI: 10.2–20.4 pp). Across all seven within-factor pairs matched within one static-accuracy point, the descriptive median RA gap is 15.1 points, and four gaps are at least 10 points. Qwen3-VL-4B provides a concrete transition-level example: on 166/200 base-correct Count triplets, the gold preference changes but its prediction does not.

RewardLens does not identify internal reasoning or replace static evaluation. It shows that a shared measured static score can leave consequential behavioral variation unresolved. Our contributions are a static-score-matched measurement framing, a programmatically constructed and validated RewardLens-CLEVR audit with relevant and irrelevant edits, and an eight-judge evaluation with paired uncertainty and pre-specified matching-band checks.

2 RELATED WORK

LLM-as-a-judge and multimodal reward evaluation. Large language models are increasingly used as automatic evaluators of generated responses. Early work established the LLM-as-a-judge paradigm and examined its agreement with human preferences, while also documenting position, verbosity, and self-enhancement biases (Zheng et al., 2023). Subsequent benchmarks evaluate judges on challenging response pairs, test-time response selection, and critique-based refinement (Tan et al., 2025; Zhou et al., 2025). These studies show that judge performance depends on the task, candidate distribution, and evaluation protocol, motivating direct evaluation of the judges rather than assuming that a capable generator is also a reliable evaluator.

This paradigm has recently been extended to multimodal inputs. MLLM-as-a-Judge evaluates multimodal judges through scoring, pairwise comparison, and batch-ranking tasks, revealing substantial variation across evaluation formats (Chen et al., 2024). MJ-Bench studies multimodal judges for text-to-image generation across dimensions including alignment, safety, image quality, bias, and composition (Chen et al., 2025), while Multimodal RewardBench 2 extends evaluation to image generation, image editing, interleaved generation, and multimodal reasoning (Hu et al., 2026). Complementary work develops increasingly general or capable multimodal reward models, including Omni-Reward, R1-Reward, and BaseReward (Jin et al., 2026; Zhang et al., 2026a;b). These evaluations primarily characterize performance on fixed examples using accuracy, ranking agreement, or correlation with human preferences. RewardLens studies a different question: whether judges assigned similar factor-specific static preference scores also behave similarly when the visual evidence governing the correct preference is changed.

Behavioral and controlled evaluation. Aggregate held-out accuracy can conceal systematic behavioral failures. CheckList formalizes behavioral testing through targeted capabilities and input transformations, including invariance tests for meaning-preserving changes and directional tests for changes that should alter a prediction (Ribeiro et al., 2020). More broadly, shortcut learning describes how models can perform well on conventional benchmarks by exploiting predictive but non-causal features that fail under carefully chosen test conditions (Geirhos et al., 2020). RewardLens adopts this behavioral perspective for multimodal judges by separately testing when a preference should change and when it should remain invariant.

Controlled contrast has a long history in vision-language evaluation. VQA v2 pairs the same question with semantically similar images that require different answers, reducing the usefulness of language-only priors and forcing predictions to depend on visual evidence (Goyal et al., 2017). VALSE uses validated foils to test grounding of existence, cardinality, attributes, actions, and spatial relations (Parcalabescu et al., 2022). Winoground, CREPE, and SugarCrepe construct contrastive examples or hard negatives to diagnose compositional vision-language understanding (Thrush et al., 2022; Ma et al., 2023; Hsieh et al., 2023). More recent work uses visual programming to generate verifiable counterfactual multimodal examples (Gao et al., 2025). These benchmarks establish the value of controlled examples for isolating particular capabilities. However, they primarily evaluate answer models or vision-language representations, rather than pairwise judges, and do not ask how much behavioral variation remains after judges are matched on an independent static preference score.

Perturbation-based audits of multimodal judges. Recent studies directly examine whether multimodal reward models and judges rely on appropriate visual evidence. Multimodal reward models can exploit unimodal, particularly text-only, spurious correlations that support in-distribution accuracy but impair out-of-distribution generalization (Li et al., 2025). Visual manipulations can also systematically inflate the scores assigned by LLM judges in text-to-image evaluation (Hwang et al., 2025). MM-JudgeBias evaluates compositional biases through controlled perturbations to the query, image, and response, distinguishing sensitivity to relevant evidence from stability under irrelevant perturbations (Lee et al., 2026). Perceptual Judgment Bias similarly studies cases in which visual evidence conflicts with plausible textual cues and proposes perturbation-based training to improve perceptual fidelity (Park et al., 2026). Multi-Crit examines another form of behavioral flexibility: whether multimodal judges can change their decisions appropriately when the requested evaluation criterion changes (Xiong et al., 2026).

VisualFLIP is the closest prior work at the intervention level: it holds the question fixed, minimally changes task-critical visual evidence, and measures whether a multimodal prediction updates when the gold answer changes (Zhu et al., 2026). RewardLens does not claim the first use of visual perturbations or prediction-update tests. It introduces a different comparison and measurement design. First, it measures factor-specific static preference accuracy on an independent natural-image set and studies the residual intervention-behavior differences among judges matched by that score. Second, each audit triplet includes both a relevant edit that reverses the gold preference and an irrelevant edit that preserves it, while holding the question, candidate responses, candidate order, camera, and rendering conditions fixed. Third, Relevant Adaptation and Irrelevant Invariance are conditioned on a correct base decision, separating selective updating from generic prediction instability. RewardLens therefore complements static judge benchmarks and prior perturbation studies; it characterizes observable intervention behavior rather than claiming to identify a judge’s internal reasoning or visual-grounding mechanism.

3 PROBLEM FORMULATION

Let a multimodal judge f receive $x = (I, q, r_1, r_2)$ and return a binary preference with gold label y . For visual factor c , its measured static accuracy on an independent set is

$$\hat{A}_{S,c}(f) = \frac{1}{n_c} \sum_{k=1}^{n_c} \mathbf{1}\{f(x_k) = y_k\}. \quad (1)$$

Claims of equality below concern this finite-sample measurement, not equality of population accuracies.

3.1 CONTROLLED TRIPLETS AND BEHAVIORAL MEASURES

An audit triplet $x_B = (I_B, q, r_1, r_2)$, $x_R = (I_R, q, r_1, r_2)$, and $x_I = (I_I, q, r_1, r_2)$ holds the linguistic preference problem fixed. Its scene contracts require $y_R \neq y_B$ for the relevant edit and $y_I = y_B$ for the irrelevant edit. For m_c triplets, RewardLens conditions on a correct base decision and estimates

$$\widehat{\text{RA}}_c(f) = \frac{\sum_k \mathbf{1}\{f(x_{B,k}) = y_{B,k} \wedge f(x_{R,k}) = y_{R,k}\}}{\sum_k \mathbf{1}\{f(x_{B,k}) = y_{B,k}\}}, \quad (2)$$

$$\widehat{\text{II}}_c(f) = \frac{\sum_k \mathbf{1}\{f(x_{B,k}) = y_{B,k} \wedge f(x_{I,k}) = y_{I,k}\}}{\sum_k \mathbf{1}\{f(x_{B,k}) = y_{B,k}\}}. \quad (3)$$

The denominator is judge-specific; Sec. 7 additionally uses the intersection of two judges’ base-correct sets.

Because the output is binary, a base-correct judge succeeds on the relevant branch exactly when it changes its prediction. Thus

$$1 - \text{RA}_c(f) = P(f(x_R) = f(x_B) \mid f(x_B) = y_B), \quad (4)$$

where prediction retention is a failure. On the irrelevant branch, retention is instead the desired behavior. RA and II therefore separate selective adaptation from generic prediction stability.

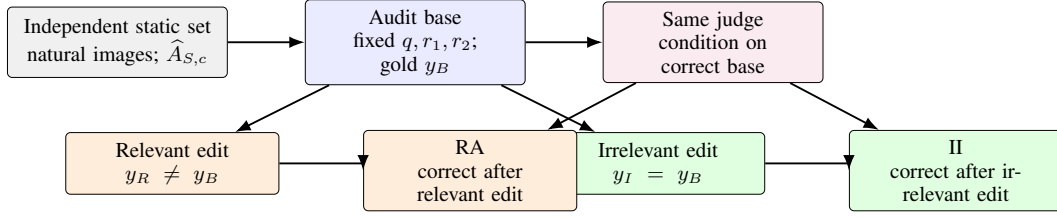


Figure 1: RewardLens separates independent static evaluation from a controlled triplet audit. Relevant edits reverse the gold preference; irrelevant edits preserve it. RA and II condition on a correct base decision.

3.2 STATIC-MATCHED MEASUREMENT QUESTION

Judges f and g are matched within factor c at tolerance ϵ when $|\hat{A}_{S,c}(f) - \hat{A}_{S,c}(g)| \leq \epsilon$. We report the residual gaps

$$\Delta_{RA,c} = |\widehat{RA}_c(f) - \widehat{RA}_c(g)|, \quad \Delta_{II,c} = |\widehat{II}_c(f) - \widehat{II}_c(g)|. \quad (5)$$

The question is empirical: whether real judges assigned equal or nearly equal measured static scores also exhibit narrow behavioral gaps under controlled evidence changes.

Proposition 1 (static-score non-identification). Without additional assumptions linking the independent static set to the controlled audit, $\hat{A}_{S,c}$ does not uniquely determine $(\widehat{RA}_c, \widehat{II}_c)$.

Proof. Fix any prediction vector on the static set. Two judges that agree on this vector have the same measured static accuracy. Because the audit contains separate inputs, the judges can retain the same static predictions while differing on audit branches: for example, one can follow the changed gold choice after every relevant edit, whereas the other retains its base choice. Their measured RA values then differ although their measured static scores are identical. The same construction applies to II by changing behavior only on the irrelevant branch. Hence the static score alone cannot recover the intervention profile. $\square$

This is a measurement statement about the two finite evaluation environments; it does not identify internal reasoning or assert that the observed judges instantiate every logically possible profile.

3.3 ANALYSIS UNITS AND INTERPRETATION

The static score and intervention profile are intentionally measured on different populations: an independent natural-image preference set and a controlled rendered audit, respectively. Thus, $\hat{A}_{S,c}$ is not the audit-base accuracy, and RA/II are not algebraic decompositions of the static score. The matching analysis is a descriptive within-factor census over the eight evaluated judges. For a matched judge pair, uncertainty is assessed by resampling common static-item identifiers and common source-triplet identifiers jointly; the resulting intervals quantify finite-evaluation contrasts, not a population-level model-family effect. Because qualifying pairs can share a judge, they are not treated as independent observations.

4 REWARDLENS AUDIT CONSTRUCTION

RewardLens-CLEVR uses CLEVR-style procedural rendering (Johnson et al., 2017) to instantiate the contracts in Sec. 3. Only the image changes within a triplet; the question, responses, candidate order, camera, lighting, and render seed are held fixed.

Factor-specific scene contracts. Both branches perform the same class of low-level operation; their difference is whether that operation changes the fact queried by the fixed preference problem.

Count. Both branches add a geometrically matched object at the same intervention slot. The relevant branch adds an object in the queried color category, whereas the irrelevant branch adds a non-target-color object.

Attribute. Both branches apply the same color transition. The relevant branch modifies the unique queried object; the irrelevant branch modifies a matched non-target object.

Presence. Both branches remove one object. The relevant branch removes the queried color–shape conjunction, whereas the irrelevant branch removes a matched non-target.

Spatial. Both branches apply the same world-space translation. The relevant branch moves the queried referent; the irrelevant branch moves a matched distractor.

Why match branches? The comparison is not an edit-versus-no-edit test. Each branch applies a matched low-level scene operation while differing in decision relevance. Relevance is defined by the deterministic gold-label contract, not by a pixel-distance threshold.

Post-render validation and scope. Factor-specific validators inspect realized scene state rather than trusting requested edits: they recount target-color objects, verify target uniqueness and color, recount the queried color–shape conjunction, or recompute the left/right relation and displacement. Every retained triplet also satisfies $y_R \neq y_B$ and $y_I = y_B$. These deterministic construction and post-render checks validate the scene contracts. Pixel change is not used to define relevance. Detailed contracts, receipts, and edit diagnostics appear in Appendix B.3.

5 EXPERIMENTAL SETUP

5.1 EVALUATION ENVIRONMENTS

The independent static evaluation contains 200 frozen items per factor (800 total). Count uses TallyQA (Acharya et al., 2019); Attribute, Presence, and Spatial use GQA (Hudson & Manning, 2019). Items pair one gold response with a structured alternative and use deterministic candidate-order shuffling. The separate RewardLens-CLEVR audit contains 200 programmatically validated triplets per factor (800 triplets; 2,400 rendered images). Static and audit examples are disjoint, and every judge receives the same items. All static estimates use 200 reported outputs except Phi-3.5-Vision on Spatial, which uses 198 reported outputs; no imputation is used.

5.2 MULTIMODAL JUDGES AND INFERENCE

We evaluate eight multimodal judges spanning multiple model families and training objectives: Qwen3-VL-4B, Gemma-3-4B, Molmo-7B-D, Skywork-VL-Reward-7B, Idefics3-8B, Phi-3.5-Vision, InternVL3-8B, and LLaVA-OneVision-7B. A common deterministic prompt requests exactly A or B; model-specific adapters parse the binary decision, while Skywork uses its reward head. Checkpoints, adapters, prompt, and decoding settings appear in Appendix B.1.

Candidate order is fixed across the three states of each triplet so that the visual input is the manipulated variable. RA and II use the judge-specific base-correct denominator in Eqs. 2–3.

5.3 STATIC MATCHING AND ANALYSIS

The frozen analysis plan specifies within-factor bands of 1, 2, and 3 percentage points. The primary rule is

$$|\hat{A}_{S,c}(f) - \hat{A}_{S,c}(g)| \leq 0.01. \quad (6)$$

All qualifying pairs are reported. Their summaries are a descriptive census: pairs can share a judge and are not independent samples. We additionally report paired-bootstrap intervals and common-base-correct comparisons. Provenance, data receipts, and commands are in Appendices B.4 and B.5.

Table 1: Complete primary $|\Delta A_S| \leq 1$ pp within-factor matched-pair census. Gaps are absolute percentage points.

Factor	Pair	ΔA_S	ΔRA	ΔII
Count	Molmo-Phi	1.00	1.37	1.91
Attribute	Phi-LLaVA	.00	15.05	10.75
Presence	Molmo-Idefics3	.50	25.63	.51
Presence	Molmo-InternVL	.50	25.63	1.01
Presence	Qwen-LLaVA	1.00	2.00	.00
Presence	Idefics3-InternVL	1.00	.00	.50
Spatial	Skywork-Phi	.35	38.60	1.56

6 RESULTS

6.1 EXACT STATIC TIE

Phi-3.5-Vision and LLaVA-OneVision-7B have identical measured static Attribute accuracy,

$$\hat{A}_{S,Attr}(\text{Phi}) = \hat{A}_{S,Attr}(\text{LLaVA}) = 80.5\%, \quad (7)$$

but RA values of 84.95% and 100.00%. Their absolute RA gap is therefore 15.05 pp (paired-bootstrap 95% CI: 10.21–20.43 pp), while their II values differ by 10.75 pp. This is a finite-evaluation separation, not a claim of equal population accuracies.

6.2 ALL ONE-POINT MATCHED PAIRS

The primary rule yields seven comparisons. Table 1 reports the full census rather than a selected subset. Their descriptive median absolute RA gap is 15.05 pp; four gaps are at least 10 pp, three exceed 20 pp, and one exceeds 30 pp. The median II gap is only 1.01 pp. Qualifying pairs can share a judge and are not treated as independent observations for inferential testing.

The census also shows that the result is neither a claim that every accuracy-matched pair has a large RA gap nor a selection of only favorable examples. The Count comparison has a 1.37 pp RA gap, and the Idefics3-InternVL Presence comparison has no observed RA gap. In contrast, the Attribute tie and the two Molmo Presence comparisons yield gaps from 15.05 to 25.63 pp, while the Spatial Skywork-Phi comparison reaches 38.60 pp. Static-score matching can therefore conceal large behavioral differences, but it does not guarantee them.

The largest gap is Spatial Skywork-Phi: 38.60 pp despite a 0.35 pp static difference. Because Spatial relevant edits have larger average pixel change than their controls, we treat this as supportive rather than the headline case. The exact Attribute tie is not explained by larger relevant edits: its average relevant pixel-change fraction is 0.118, versus 0.179 for irrelevant edits.

6.3 BEHAVIORAL PROFILES AND PREDICTION TRANSITIONS

Across all 32 cells, RA ranges from 0 to 1, whereas II ranges from about 0.876 to 1 (Fig. 2). High invariance therefore does not imply appropriate adaptation. Qwen illustrates the distinction: it has $RA/II = 1/1$ on Attribute but $RA/II = .17/1$ on Count. It is correct on every Count base image and follows the new preference on only 34/200 relevant edits. In the remaining 166/200 cases, it retains its base prediction even though the gold label changes, exactly the failure in Eq. 4. Static accuracy is informative, but it does not determine these selective response patterns.

7 ROBUSTNESS AND VALIDITY CHECKS

7.1 SHARED-BASE-CORRECT COMPARISONS

The Attribute tie is unchanged on the 186 triplets that both Phi and LLaVA judge correctly at base: RA remains 84.95%/100.00%. Across all seven one-point matched pairs, restricting comparisons to shared base-correct support leaves the descriptive median absolute RA gap at 15.05 pp. This

Relevant Adaptation (RA)					Irrelevant Invariance (II)				
	Count	Attribute	Presence	Spatial		Count	Attribute	Presence	Spatial
Qwen	0.17	1.00	1.00	0.98	Qwen	1.00	1.00	1.00	1.00
Gemma	0.32	0.95	0.60		Gemma	0.88	0.97	1.00	0.98
Molmo	0.19	0.99	0.74	0.40	Molmo	0.97	0.99	0.99	0.99
Skywork	0.63	1.00	1.00	0.86	Skywork	0.99	1.00	0.98	1.00
Idefics3	0.71	1.00	1.00	0.55	Idefics3	0.96	1.00	0.99	1.00
Phi	0.20	0.85	0.74	0.47	Phi	0.95	0.89	0.97	0.98
LLaVA	0.42	1.00	0.98	0.87	LLaVA	0.99	1.00	1.00	1.00
InternVL	0.18	1.00	1.00	0.92	InternVL	1.00	1.00	1.00	1.00

Figure 2: All $8 \times 4 = 32$ model-factor measurements. Relevant Adaptation (RA) varies widely; Irrelevant Invariance (II) is predominantly high. Both are conditional on a correct audit-base decision.

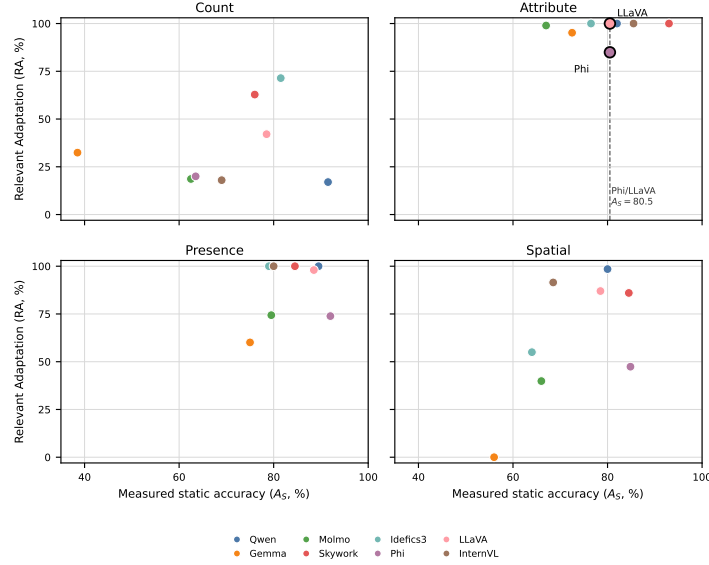


Figure 3: Measured static accuracy versus RA by factor. The Attribute panel highlights the Phi-LLaVA exact tie at 80.5% static accuracy and their 84.95% versus 100.00% RA values.

diagnostic uses a common support for both judges; it complements rather than replaces the judge-conditional RA definition.

7.2 AUDIT-BASE ACCURACY

The matched static comparison is not simply an artifact of a judge being wrong on most audit bases. For example, on Presence, Molmo and Idefics3 have audit-base accuracies of 99.5% and 100.0%, respectively, but differ by 25.63 pp in RA. The full audit-base ledger and judge-specific conditional denominators are reported in Table 4 in Appendix A.

7.3 MATCHING-THRESHOLD SENSITIVITY

The descriptive median absolute RA gap is 15.05 pp under the 1 pp matching band, 15.05 pp under the 2 pp band, and 15.14 pp under the 3 pp band. These bands are part of the frozen analysis plan; the complete censuses are reported in Table 7 in Appendix A. Pair rows share judges and are not treated as independent observations.

7.4 MISSING-OUTPUT SENSITIVITY

Phi-3.5-Vision has 198 reported Spatial static outputs, with 168 correct. Its reported estimate is therefore $168/198 = 84.85\%$. A conservative boundary check treats both unavailable outcomes as errors or both as correct, yielding $168/200 = 84.00\%$ and $170/200 = 85.00\%$, respectively. Either

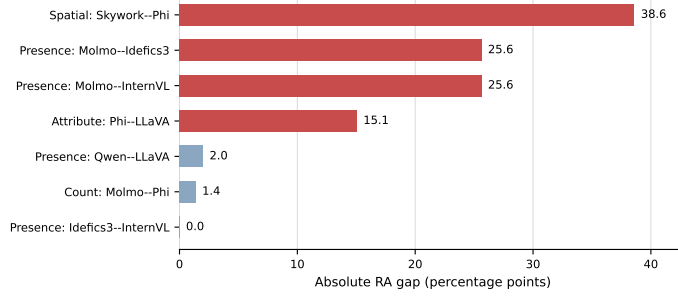


Figure 4: Ranked RA gaps for all seven primary matched pairs. Red bars mark gaps of at least 10 pp.

endpoint remains within 1 pp of Skywork’s 84.50% Spatial static accuracy, so the Skywork-Phi pair qualifies under the primary matching rule throughout this entire worst-to-best-case range. We do not attribute a cause to the two unavailable outcomes because the frozen receipts do not establish one.

7.5 EDIT-MAGNITUDE SENSITIVITY AND SCOPE

On Attribute, relevant edits have lower average pixel-change fraction than irrelevant edits (0.118 versus 0.179). In the post-hoc Attribute subset with matched edit magnitude, the Phi-LLaVA RA gap remains 16.8 pp. Spatial has a larger relevant-edit magnitude, so its 38.60 pp Skywork-Phi gap remains supportive rather than the headline case. The complete common-support, paired-bootstrap, and edit-magnitude checks are reported in Appendix A.2. These checks support a limited behavioral conclusion, not an internal-grounding claim.

8 DISCUSSION

What a static tie does and does not establish. Static preference accuracy remains useful: it measures correctness on a fixed, independent evaluation distribution. However, it is a many-to-one summary of behavior. The Attribute Phi-LLaVA tie and the complete matched-pair census show that a shared measured score need not determine how a judge responds when the visual fact supporting a preference changes. RewardLens is therefore a complement to, rather than a replacement for, static evaluation: it asks for a second measurement axis when behavioral response to visual changes matters.

Adaptation is distinct from generic stability. The two intervention branches prevent a simple preference for either changing or retaining a prediction. On a relevant edit, retention by a base-correct judge is an error because the gold preference changes; on an irrelevant edit, retention is the desired response. Qwen’s Count behavior makes this distinction concrete: its high II does not offset its low RA. Reporting both quantities therefore distinguishes selective adaptation from generic prediction stability.

Limitations and scope. RewardLens measures observable behavior under four controlled visual factors; it does not identify a model’s internal reasoning process or establish causal claims about deployed-system outcomes. The results should consequently be interpreted as evidence that static scores can leave factor-specific intervention behavior unresolved in this evaluation setting, not as a universal ranking of multimodal judges. Programmatic checks validate the rendered scene contracts but are not a substitute for human perceptual validation. Candidate order is held fixed within each triplet; because an orientation-swap audit is not part of the reported experiment, RA and II are estimates under this frozen pair orientation rather than position-bias-free quantities.

9 CONCLUSION

RewardLens shows that equal or near-equal measured static accuracy can coexist with different measured intervention behavior. The clearest case is the 80.5% Attribute tie between Phi and LLaVA, whose RA differs by 15.1 pp; the complete one-point census has seven pairs and a descriptive median RA gap of 15.1 pp. These results complement rather than invalidate static evaluation. They apply to observable behavior on four controlled factors and do not identify internal grounding mechanisms. Within that scope, reporting RA and II alongside static accuracy exposes selective failures to update that a single cross-sectional score leaves unresolved.

AI USE STATEMENT

Generative AI tools assisted with manuscript organization and language editing, $\text{\LaTeX}$ formatting, code and data-analysis support, and feedback on the presentation of the methodology and empirical results. The authors designed the study, reviewed and verified all scientific claims, experimental results, and citations, and take full responsibility for the final manuscript.

ETHICS STATEMENT

RewardLens evaluates publicly available model checkpoints on existing benchmark material and procedurally rendered scenes. It does not involve human subjects or collect personal data. The reported behavioral measurements should not be interpreted as claims about model internals or deployed-system safety.

REPRODUCIBILITY STATEMENT

Sections 4 and 5 specify the triplet construction and validation rules, data sources and sample sizes, models, evaluation protocol, and static-matching analysis used for the reported measurements.

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A ADDITIONAL RESULTS

This appendix reports the complete factor-level result ledger and the full robustness summaries used in the main text. Percentages are reported to two decimal places. A_S is measured on the independent static set; A_B is unconditioned accuracy on the RewardLens-CLEVR base images; RA and II are conditional on a correct base decision.

A.1 ALL MODEL-FACTOR MEASUREMENTS

Table 2 contains every one of the $8 \times 4 = 32$ model-factor measurements. Each factor has 200 frozen static items and 200 frozen audit triplets. All static estimates use 200 reported outcomes except Phi-3.5-Vision on Spatial ($n = 198$); no imputation is used. The conditional denominator n_B is the number of base-correct audit triplets for that model and factor.

A.2 FULL ROBUSTNESS TABLES

Paired bootstrap. For each primary matched pair, we resample matched source triplets jointly for the two judges (10,000 replicates; seed 20270915). Table 5 reports signed model-A minus model-B contrasts, so its sign depends on the printed order. Static contrasts are resampled on shared static item identifiers; the reported static point estimate can therefore differ slightly from the displayed difference between separately rounded static percentages.

Shared-base-correct support. Table 6 recomputes RA and II on the intersection of the two judges’ base-correct sets. It is a common-support diagnostic, not a replacement for the judge-conditional definitions in Eqs. 2–3.

As an additional post-hoc sensitivity analysis, the magnitude-matched subset retains 459 of 800 triplets under $|p_R - p_I| \leq 0.05$: 191 Count, 109 Attribute, 103 Presence, and 56 Spatial triplets. On the 109 retained Attribute triplets, Phi is base-correct on 101 and has RA = 83.17%; LLaVA has RA = 100.00%. This is an approximately 16.8 percentage-point RA gap on that restricted subset.

A.3 REPRESENTATIVE FROZEN AUDIT TRIPLETS

Figure 5 shows one retained PASS triplet per factor. These are direct copies of frozen renders; no scene was regenerated or manually edited for the manuscript. Count changes or preserves the target-color count; Attribute recolors the target or a matched non-target; Presence removes the queried conjunction or a matched non-target; and Spatial moves the queried referent or a matched distractor.

Table 2: Complete model-factor ledger. Each cell is A_S /RA/II in percent; the corresponding audit-base denominators are reported in Table 4. All A_S estimates use $n = 200$ except Phi-3.5-Vision on Spatial ($n = 198$; marked †); no imputation is used.

Judge	Count			Attribute			Presence			Spatial		
	A_S	RA	II	A_S	RA	II	A_S	RA	II	A_S	RA	II
Qwen3-VL-4B	91.50	17.00	100.00	82.00	100.00	100.00	89.50	100.00	100.00	80.00	98.50	100.00
Gemma-3-4B	38.50	32.41	87.59	72.50	95.19	97.33	75.00	60.10	100.00	56.00	0.00	97.95
Molmo-7B-D	62.50	18.63	97.06	67.00	98.96	98.96	79.50	74.37	98.99	66.00	39.86	99.32
Skywork-VL-Reward-7B	76.00	62.81	98.99	93.00	100.00	100.00	84.50	100.00	98.49	84.50	86.00	100.00
Idefics3-8B	81.50	71.43	95.77	76.50	100.00	100.00	79.00	100.00	99.50	64.00	55.00	100.00
Phi-3.5-Vision	63.50	20.00	95.15	80.50	84.95	89.25	92.00	73.89	97.22	84.85 †	47.40	98.44
LLaVA-OneVision-7B	78.50	42.08	98.91	80.50	100.00	100.00	88.50	98.00	100.00	78.50	87.00	100.00
InternVL3-8B	69.00	18.00	100.00	85.50	100.00	100.00	80.00	100.00	100.00	68.50	91.50	100.00

Table 3: Complete $|\Delta A_S| \leq 1$ pp matched-pair census. All gaps are absolute percentage points.

Factor	Pair	ΔA_S	ΔRA	ΔII
Spatial	Skywork-Phi	.35	38.60	1.56
Presence	Molmo-Idefics3	.50	25.63	.51
Presence	Molmo-InternVL	.50	25.63	1.01
Attribute	Phi-LLaVA	.00	15.05	10.75
Presence	Qwen-LLaVA	1.00	2.00	.00
Count	Molmo-Phi	1.00	1.37	1.91
Presence	Idefics3-InternVL	1.00	.00	.50

Table 4: Audit-base accuracy and conditional denominators for the same 32 measurements in Table 2. Each entry is A_B (%) / n_B .

Judge	Count	Attribute	Presence	Spatial
Qwen3-VL-4B	100.0 / 200	100.0 / 200	99.5 / 199	100.0 / 200
Gemma-3-4B	72.5 / 145	93.5 / 187	99.0 / 198	73.0 / 146
Molmo-7B-D	51.0 / 102	96.5 / 193	99.5 / 199	74.0 / 148
Skywork-VL-Reward-7B	99.5 / 199	100.0 / 200	99.5 / 199	100.0 / 200
Idefics3-8B	94.5 / 189	100.0 / 200	100.0 / 200	100.0 / 200
Phi-3.5-Vision	82.5 / 165	93.0 / 186	90.0 / 180	96.0 / 192
LLaVA-OneVision-7B	91.5 / 183	100.0 / 200	100.0 / 200	100.0 / 200
InternVL3-8B	100.0 / 200	100.0 / 200	100.0 / 200	100.0 / 200

Table 5: Paired-bootstrap contrasts for all primary matched pairs (percentage points; 95% percentile intervals). All audit contrasts resample 200 matched source triplets jointly across the two judges.

Factor	Ordered comparison	Static Δ [95% CI]	RA Δ [95% CI]	II Δ [95% CI]
Spatial	Skywork — Phi	−0.51 [−6.06, 5.05]	38.60 [29.78, 47.23]	1.56 [0.00, 3.61]
Attribute	Phi — LLaVA	0.00 [−5.50, 5.50]	−15.05 [−20.43, −10.21]	−10.75 [−15.46, −6.42]
Presence	Molmo — Idefics3	0.50 [−6.50, 7.50]	−25.63 [−31.66, −19.60]	−0.51 [−2.04, 1.00]
Presence	Molmo — InternVL	−0.50 [−6.00, 5.00]	−25.63 [−31.82, −19.60]	−1.01 [−2.53, 0.00]
Count	Molmo — Phi	−1.00 [−6.50, 4.50]	−1.37 [−10.23, 7.55]	1.91 [−2.68, 6.40]
Presence	Qwen — LLaVA	1.00 [−3.00, 5.00]	2.00 [0.50, 4.00]	0.00 [0.00, 0.00]
Presence	Idefics3 — InternVL	−1.00 [−7.00, 5.00]	0.00 [0.00, 0.00]	−0.50 [−1.50, 0.00]

Table 6: All seven primary matched pairs on shared base-correct support. Entries show model-A/model-B percentages; gaps are absolute percentage points.

Factor	Pair	$n_\cap$	RA A/B	$ \Delta RA $	II A/B	$ \Delta II $
Spatial	Skywork-Phi	192	86.98 / 47.40	39.58	100.00 / 98.44	1.56
Presence	Molmo-Idefics3	199	74.37 / 100.00	25.63	98.99 / 99.50	0.50
Presence	Molmo-InternVL	199	74.37 / 100.00	25.63	98.99 / 100.00	1.01
Attribute	Phi-LLaVA	186	84.95 / 100.00	15.05	89.25 / 100.00	10.75
Count	Molmo-Phi	100	17.00 / 4.00	13.00	99.00 / 100.00	1.00
Presence	Qwen-LLaVA	199	100.00 / 97.99	2.01	100.00 / 100.00	0.00
Presence	Idefics3-InternVL	200	100.00 / 100.00	0.00	99.50 / 100.00	0.50

Table 7: Sensitivity to the within-factor static-matching tolerance. All gaps are absolute percentage points.

Tolerance	Pairs	Median RA	90th RA	Max RA	RA $\geq$ 10 pp
1 pp	7	15.05	30.82	38.60	4
2 pp	11	15.05	25.63	38.60	7
3 pp	15	15.14	34.90	51.64	11

Table 8: Image-level edit diagnostics by factor. Pixel-change fraction and global SSIM are computed after resizing to 256 pixels; SSIM is a global RGB statistic, not a semantic mask measurement.

Factor	Pixel-change fraction		Global SSIM	
	Relevant	Irrelevant	Relevant	Irrelevant
Count	0.229	0.229	0.925	0.922
Attribute	0.118	0.179	0.974	0.958
Presence	0.175	0.201	0.935	0.918
Spatial	0.375	0.265	0.817	0.890

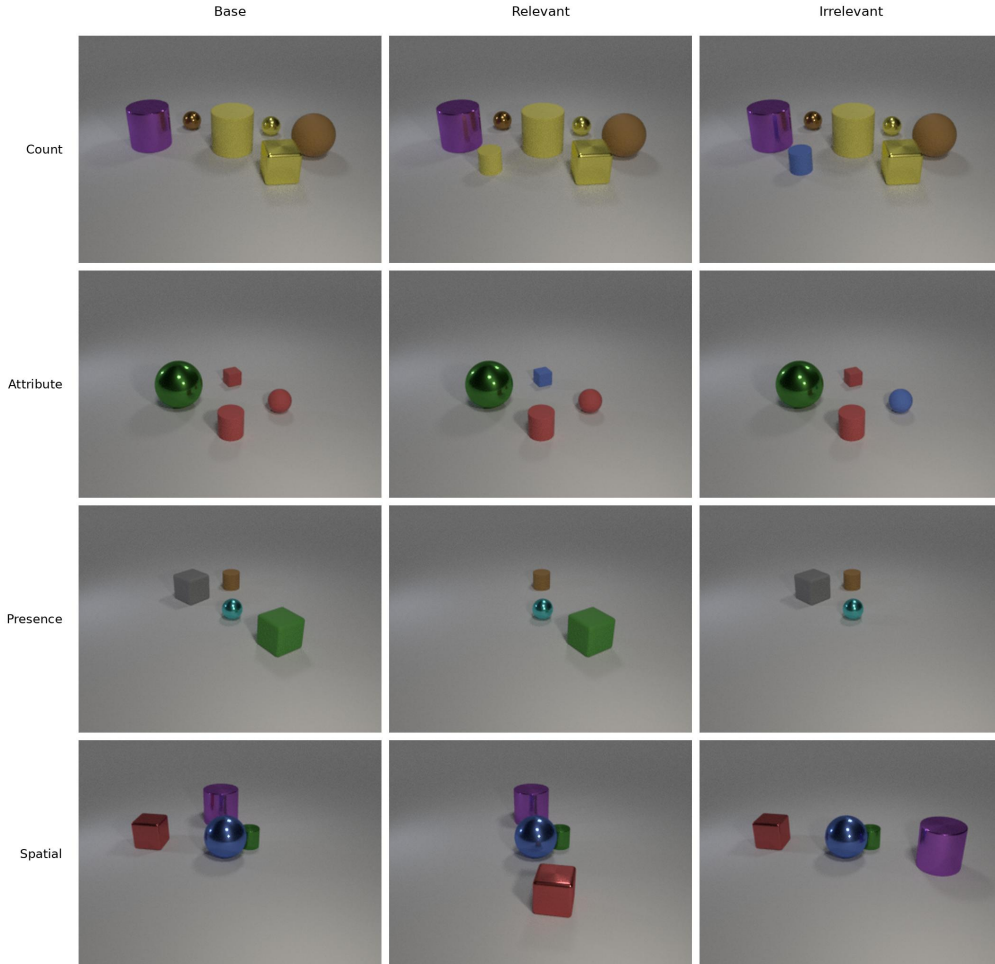


Figure 5: Representative frozen RewardLens-CLEVR PASS triplets. Columns are Base, Relevant, and Irrelevant. Rows respectively illustrate target-category count change versus a matched non-target count change; target versus non-target recoloring; queried-conjunction versus matched non-target removal; and queried referent versus matched distractor translation. The source triplet identifiers are Count count_000000, Attribute attr_000000, Presence pres_000000, and Spatial spat_000000.

B IMPLEMENTATION DETAILS

B.1 CHECKPOINTS, PROMPT, AND DECODING

Table 9 lists the eight evaluated checkpoints. The frozen run receipts retain the resolved revision and checkpoint-file hashes; the table uses the public repository identifier so that the model identity is auditable without embedding machine-local paths.

Every non-reward judge received the same frozen prompt, with the image supplied through the model’s chat-template image placeholder:

```
You are scoring a pairwise visual
question-answering preference.
Question: {question}
Candidate A: {candidate_a}
Candidate B: {candidate_b}
Look at the image. Choose the better
answer to the question.
Reply with exactly one token: A or B.
```

The parser accepts the binary A/B decision. Generation is deterministic (`do_sample = False`, temperature unset, and at most 16 new tokens). The expanded checkpoint adapter uses bfloat16 and SDPA; Phi uses eager attention because its frozen environment required separate handling. The Skywork adapter scores the two candidates with its value head rather than using generative A/B output. Candidate order is fixed within an item across judges and deterministically shuffled across items.

B.2 STATIC PREFERENCE SET CONSTRUCTION

The independent static set has 800 natural-image preference problems: 200 per factor. Count is drawn from TallyQA (Acharya et al., 2019) and contains a gold count paired with a nearby incorrect count. Attribute, Presence, and Spatial are drawn from GQA (Hudson & Manning, 2019); their alternatives use, respectively, structured attribute substitutions, presence alternatives, and relation inversions or binary alternatives. Each item stores a single gold preference after a deterministic A/B order shuffle.

Static and audit images are disjoint by construction. The RewardLens-CLEVR audit is a separate procedurally rendered environment and shares neither the natural-image examples nor their selection procedure.

B.3 REWARDLENS-CLEVR GENERATION AND VALIDATION

The audit is frozen at 200 PASS triplets per factor. It uses CLEVR-style procedural scenes (Johnson et al., 2017). A triplet contains three renders—base, relevant, and irrelevant—for 800 triplets and 2,400 images in total. The retained set is the first 200 sorted PASS identifiers per factor; borderline or failed triplets are not promoted.

The construction and validator implement the contracts summarized in Sec. 4. Count validates target-color recounting after matched object additions. Attribute validates target uniqueness and target or non-target recoloring. Presence validates the queried color-shape conjunction after matched removals. Spatial recomputes the queried left/right relation from the rendered 3D scene state and checks the matched world-space displacement. All validators also enforce the gold-label contract ($y_R \neq y_B$ and $y_I = y_B$), as well as shared camera, lighting, and render seed within a triplet. These programmatic checks validate the rendered scene specification.

B.4 ANALYSIS-PLAN PROVENANCE AND DATA RECEIPTS

Analysis-plan provenance. The retained version-controlled analysis plan specifies the 1, 2, and 3 percentage-point within-factor accuracy-matching bands. Identifying repository metadata, timestamps, and cryptographic receipt digests are omitted during double-blind review. We refer to this document as a *frozen analysis plan*; it is not a formal preregistration.

Table 9: Evaluated checkpoints and inference adapter. All checkpoints were evaluated as binary pairwise judges; Skywork uses its reward adapter.

Display name	Checkpoint repository	Adapter
Qwen3-VL-4B	Qwen/Qwen3-VL-4B-Instruct	Qwen pairwise
Gemma-3-4B	google/gemma-3-4b-it	Transformers pairwise
Molmo-7B-D	allenai/Molmo-7B-D-0924	Molmo native pairwise
Skywork-VL-Reward-7B	Skywork/Skywork-VL-Reward-7B	Skywork reward head
Idefics3-8B	AI-ModelScope/Idefics3-8B-Llama3	frozen expanded pairwise
Phi-3.5-Vision	LLM-Research/Phi-3.5-vision-instruct	frozen expanded pairwise
LLaVA-OneVision-7B	AI-ModelScope/llava-onevision-qwen2-7b-ov-hf	frozen expanded pairwise
InternVL3-8B	OpenGVLab/InternVL3-8B-hf	frozen expanded pairwise

Data and evaluation receipts. The frozen audit manifest retains 200 validated triplets per factor: 800 triplets and 2,400 rendered images in total. The independent static evaluation contains 200 items per factor (800 total): Count comes from TallyQA, and Attribute, Presence, and Spatial come from GQA. The manifest and integrity receipt are retained in the project archive; identifying digests are omitted during double-blind review.

B.5 REPRODUCIBILITY COMMANDS

For double-blind review, exact local paths and command names are omitted. The project archive retains programmatic scene generation and validation, manifest freezing, offline inference preflight, metric recomputation over frozen judgments, paired bootstrap, and manuscript compilation. The paired-bootstrap artifact uses 10,000 resamples with seed 20270915 and resamples both models on the same source-triplet keys. These materials are retained for the archival release package.

B.6 RELATED-WORK SCOPE

RewardLens is not presented as the first use of visual perturbations or of prediction-update tests. VisualFLIP changes task-critical visual evidence while holding the question fixed and evaluates prediction updates (Zhu et al., 2026); MM-JudgeBias and Perceptual Judgment Bias study controlled multimodal-judge perturbations and perceptual bias (Lee et al., 2026; Park et al., 2026). RewardLens adds a different comparison: it first measures static preference accuracy on a separate, factor-specific natural-image set, then reports the remaining RA and II gaps among judges matched by that measured static score. The relevant and irrelevant branches share a fixed question, candidate pair, and candidate order within an audit triplet and condition RA/II on a correct base decision; this framing does not invalidate earlier intervention work or claim to observe internal grounding.